ISOPERIMETRIC BALANCE ON ONE PAGE

Two ratios, one best rectangle

A product of shares connects fixed perimeter, three classical means, cos and cosh.

The combination: multiply the two shares

For $a, b > 0$, take their shares of the same total:

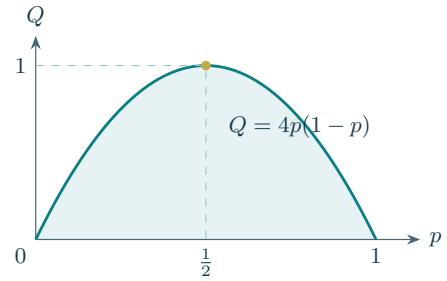
$$p = \frac{a}{a+b}, \qquad q = \frac{b}{a+b}, \qquad p + q = 1.$$

Multiply and normalise the product to have maximum 1:

$$\boxed{Q = 4pq = \frac{4ab}{(a+b)^2} = 1 - (p-q)^2 \le 1.}$$

Maximum at $a = b$; the loss $1 - Q$ is squared imbalance.

Balance wins



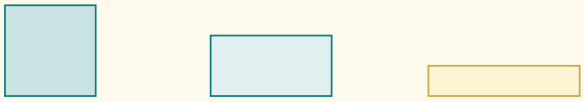
Geometry: fix the perimeter, maximise the area

A rectangle has perimeter $P = 2(a+b)$ and area $S = ab$. Thus

$$Q = \frac{16S}{P^2} \le 1, \qquad S \le \frac{P^2}{16}.$$

Equality gives the square: $a = b = P/4$.

Each rectangle has $P = 12$.



$(a,b) = (3,3)$, $S = 9$, $Q = 1$ | $(4,2)$, $S = 8$, $Q = \frac{8}{9}$ | $(5,1)$, $S = 5$, $Q = \frac{5}{9}$

The same score, written through the three means

$$A = \frac{a+b}{2}, \qquad G = \sqrt{ab}, \qquad H = \frac{2ab}{a+b}.$$

$$\boxed{Q = \frac{H}{A} = \left(\frac{G}{A}\right)^2 = \frac{16S}{P^2}, \qquad AH = G^2.}$$

Hyperbolic: positive ratios

For $x = \frac{1}{2}\ln(a/b)$, write $a = \ell e^{x}$, $b = \ell e^{-x}$, with $\ell = \sqrt{ab}$.

$$A = \ell \cosh x, \quad G = \ell, \quad H = \ell \operatorname{sech} x.$$

The imbalance is $p - q = \tanh x$, so

$$\boxed{Q = \operatorname{sech}^2 x = 1 - \tanh^2 x.}$$

At fixed P, the scale adjusts to $\ell = P/(4\cosh x)$:

$$S = \frac{P^2}{16}\operatorname{sech}^2 x.$$

Circular: the same algebra

Use the complex pair $a = \ell e^{ix}$, $b = \ell e^{-ix}$, with real x and $\ell > 0$. Formally,

$$A = \ell \cos x, \quad G = \ell, \quad H = \ell \sec x.$$

Now $p - q = i \tan x$. The identity $Q = 1 - (p-q)^2$ gives

$$\boxed{Q = \sec^2 x = 1 + \tan^2 x.}$$

For these complex shares, $Q \le 1$ need not hold. This is an algebraic extension, defined when $\cos x \neq 0$.

If the two contributions carry different weights

For positive weights α, β, fix the weighted budget $C = \alpha a + \beta b$ and use

$$p_w = \frac{\alpha a}{C}, \qquad q_w = \frac{\beta b}{C}, \qquad Q_w = 4p_w q_w = \frac{4\alpha\beta S}{C^2} \le 1.$$

Thus $S_{\max} = \dfrac{C^2}{4\alpha\beta}$, attained when $\boxed{\alpha a = \beta b}$, or $a/b = \beta/\alpha$. The optimum balances the **weighted contributions**.

Normalisation. $Q = 16S/P^2$ is normalised to the best rectangle, the square. The usual planar quotient is $4\pi S/P^2 = (\pi/4)Q$, so it equals $\pi/4$ for a square. Except in the circular panel, all lengths and weights are positive real numbers.